PowerSense: Autonomous SEL Protection System for CubeSat Power Regulators

Abstract

CubeSat missions face critical failure risks from Single Event Latch-up (SEL) in power regulators caused by space radiation. Traditional threshold-based protection is too slow, allowing permanent thermal damage. PowerSense deploys AI-powered monitoring that detects SEL patterns within 5ms and autonomously triggers tiered responses, preventing destruction and extending mission lifetime.

Single Event Latch-up in Power Regulators

Space radiation induces short circuits in voltage regulator ICs, causing excessive current (5–10× normal) and rapid thermal runaway. Without fast intervention, temperatures can exceed 300°C within 50–100ms, causing silicon destruction. Threshold-based systems are too slow to prevent permanent damage.

Mission Impact

- 70% of SEL events cause permanent damage with delayed detection
- 500ms response time of traditional systems vs. 10ms with AI
- 30% of CubeSat mission failures are due to power system faults
- Current approaches lack pattern recognition for early detection

SOLUTION

We developed an AI-based monitoring system using Isolation Forests to detect subtle deviations before failure. Detected anomalies trigger autonomous actions (load-shedding, safe-modes).

The project combines real-time current monitoring with ML pattern recognition to detect SEL signatures within milliseconds. The response strategy:

- Tier 1: Immediate power cycling (10ms) for confirmed SEL events.
- Tier 2: Autonomous reconfiguration to redundant systems for persistent faults.
- AI engine: Isolation Forest and pattern recognition differentiate SEL from transients.

WHY VOLTAGE REGULATORS?

Voltage regulators supply stable power rails for all spacecraft systems; their failure causes total loss. Early anomaly detection enables predictive maintenance and prevents mission loss.

WHY MACHINE LEARNING?

Thresholds only catch obvious faults. ML reveals complex, multi-parameter degradation patterns:

- Rapid current rise with temperature spike (SEL onset)
- Increasing voltage ripple (capacitor stress)
- Efficiency loss under thermal load (MOSFET degradation)

OUR APPROACH

Isolation Forest is trained on healthy data, flagging deviations as anomalies. This unsupervised method works with limited failure data and detects unknown degradation or SEL onset.

HARDWARE

Target (flight-grade) components:

- NanoMind A3200—onboard AI processing.
- ADS127L01-SP—precision ADC.
- LEM HTFS-SP sensors—power monitoring.
- TMP100-SP—temperature feedback.
- Vishay STHV relays—autonomous load control.
- ATmegaS128—watchdog/safety supervisor.

Prototype (COTS substitutes):

- Raspberry Pi 4 as processor.
- INA219/INA226 for current/voltage.
- MCP9808 + NTC sensors for temperature.
- Commercial ADCs/relays in place of space-grade parts.
- ESP32 as watchdog (ATmegaS128 unavailable).

Note: COTS alternatives validated the concept and model.

Space environment readiness

Space-optimized hardware meets 3U CubeSat constraints: NanoMind A3200 processor (1.5W, rad-hard), space-grade sensors, and solid-state relays autonomously monitor voltage regulation within 0.5U and 2.1W power budget, ensuring reliability in radiation, thermal cycling, and vacuum.

RESULTS

Classification Report:

	Precision	Recall	F1-Score	Support
Normal (0)	0.98	0.99	0.99	9600
Anomaly (1)	0.78	0.43	0.55	398
Accuracy	0.97			
Macro Avg	0.88	0.71	0.77	9998
Weighted Avg	0.97	0.97	0.97	9998

Model Performance (Summary):

- Accuracy: 97.00%
- Precision (Anomaly): 78.00%
- Recall (Anomaly): 43.00%
- F1-Score (Anomaly): 55.00%

Detection Example:

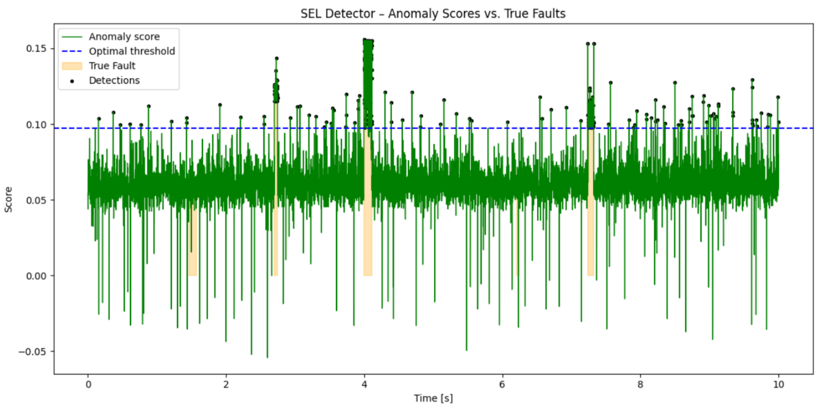

Figure 1: Anomaly score plot showing early SEL detection within 5ms window.

CONCLUSION

PowerSense AI detects early SEL and degradation in voltage regulators and executes automatic, protective actions within milliseconds. COTS prototypes prove real-time detection and readiness for CubeSat integration with radiation-tolerant hardware.